

Question 1

Thought Process

To find the ratio of maximum to full-load torque, I need to calculate the full-load slip (s_{fl}) and the slip at maximum torque (s_{mT}). The synchronous speed (N_s) can be found using the frequency and number of poles. Once s_{fl} and s_{mT} are known, the standard torque ratio formula $\frac{T_{max}}{T_{fl}} = \frac{s_{fl}^2 + s_{mT}^2}{2 \cdot s_{fl} \cdot s_{mT}}$ can be applied. The speed at maximum torque is derived directly from s_{mT} . For maximum starting torque, the slip at maximum torque must be shifted to 1, which means the new total rotor resistance must equal the standstill rotor reactance.

Answer:

Given Data:

- Power Output $P = 746$ kW[cite: 6]
- Phases = 3[cite: 6]
- Frequency $f = 50$ Hz[cite: 6]
- Number of poles $P = 16$ [cite: 6]
- Rotor resistance $R_2 = 0.02 \Omega$ [cite: 6]
- Rotor reactance $X_2 = 0.15 \Omega$ [cite: 6]
- Full load speed $N_{fl} = 360$ rpm[cite: 6]

(i) Ratio of maximum to full load torque (T_{max}/T_{fl}):

First, calculate the synchronous speed (N_s):

$$N_s = \frac{120 \cdot f}{P} = \frac{120 \cdot 50}{16} = 375 \text{ rpm}$$

Next, calculate the full-load slip (s_{fl}):

$$s_{fl} = \frac{N_s - N_{fl}}{N_s} = \frac{375 - 360}{375} = 0.04$$

Calculate the slip at maximum torque (s_{mT}):

$$s_{mT} = \frac{R_2}{X_2} = \frac{0.02}{0.15} = 0.1333$$

Using the torque ratio formula:

$$\frac{T_{max}}{T_{fl}} = \frac{s_{fl}^2 + s_{mT}^2}{2 \cdot s_{fl} \cdot s_{mT}} = \frac{0.04^2 + 0.1333^2}{2 \cdot 0.04 \cdot 0.1333} = \frac{0.0016 + 0.01777}{0.010664} \approx 1.816$$

(ii) Speed at maximum torque (N_{mT}):

$$N_{mT} = N_s(1 - s_{mT}) = 375(1 - 0.1333) = 325 \text{ rpm}$$

(iii) Rotor resistance to be added to get maximum starting torque:

To achieve maximum torque at starting, the new slip at maximum torque must be 1 ($s_{mT,new} = 1$).

$$s_{mT,new} = \frac{R_{2,total}}{X_2} \implies 1 = \frac{R_2 + R_{ext}}{X_2}$$

$$R_{ext} = X_2 - R_2 = 0.15 - 0.02 = 0.13 \Omega$$

An external resistance of 0.13Ω per phase must be added.

Question 2

Thought Process

I am given the maximum torque, the speed at which it occurs, and the rotor resistance. From the speed at maximum torque, I can find the slip at maximum torque (s_{mT}). Knowing s_{mT} and R_2 , I can calculate the rotor standstill reactance (X_2). The problem asks for the external resistance required to make the starting torque equal to half of the maximum torque. I will use the torque ratio equation $\frac{T_{st}}{T_{max}} = \frac{2a}{1+a^2}$ where $a = \frac{R_{total}}{X_2}$, set it to 0.5, and solve the resulting quadratic equation for the new resistance.

Answer:

Given Data:

- Number of poles $P = 4$ [cite: 6]
- Frequency $f = 50$ Hz[cite: 6]
- Phases = 3[cite: 6]
- Maximum torque $T_{max} = 162.8$ N-m[cite: 6]
- Speed at max torque $N_{mT} = 1365$ rpm[cite: 6]
- Rotor resistance $R_2 = 0.2$ Ω /phase[cite: 6]

Calculate the synchronous speed (N_s):

$$N_s = \frac{120 \cdot 50}{4} = 1500 \text{ rpm}$$

Calculate the slip at maximum torque (s_{mT}):

$$s_{mT} = \frac{N_s - N_{mT}}{N_s} = \frac{1500 - 1365}{1500} = 0.09$$

Determine the rotor reactance (X_2):

$$X_2 = \frac{R_2}{s_{mT}} = \frac{0.2}{0.09} = 2.222 \text{ } \Omega$$

We require the starting torque to be half of the maximum torque:

$$\frac{T_{st}}{T_{max}} = 0.5$$

[cite: 6] Using the relation for starting torque:

$$\frac{T_{st}}{T_{max}} = \frac{2a}{1+a^2} \quad \text{where } a = \frac{R_{new}}{X_2}$$

$$0.5 = \frac{2a}{1+a^2} \implies 1+a^2 = 4a \implies a^2 - 4a + 1 = 0$$

Solving the quadratic equation for a :

$$a = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3}$$

This gives two values: $a_1 = 3.732$ and $a_2 = 0.268$.

Calculate the new total rotor resistance for both cases:

$$R_{new1} = 3.732 \cdot 2.222 = 8.293 \text{ } \Omega$$

$$R_{new2} = 0.268 \cdot 2.222 = 0.595 \text{ } \Omega$$

Calculate the external resistance to be added ($R_{ext} = R_{new} - R_2$):

$$R_{ext1} = 8.293 - 0.2 = 8.093 \, \Omega$$

$$R_{ext2} = 0.595 - 0.2 = 0.395 \, \Omega$$

To limit high starting currents effectively, the higher resistance value is practically chosen. Thus, $8.093 \, \Omega$ must be inserted in series with each rotor phase.

Question 3

Thought Process

A 4-pole 50Hz motor has a synchronous speed of 1500 rpm. The provided full-load speed of 125 rpm represents an extreme slip ($> 91\%$), indicating a stalled or plugging state, which contradicts standard full-load behavior. It is a highly probable typographical error in the source text for 1425 rpm. To yield physically meaningful motor calculations, I will proceed by correcting the assumed typo to 1425 rpm. First, I will calculate the stator phase voltage and refer it to the rotor using the turns ratio. Then, I will compute the rotor current at the operating slip, followed by copper loss, air-gap power, developed mechanical power, and output power. Finally, the standard formulas for maximum and starting torques will be applied.

Answer:

Given Data:

- Voltage $V_L = 440 \, \text{V}$ [cite: 6]
- Frequency $f = 50 \, \text{Hz}$ [cite: 6]
- Number of poles $P = 4$ [cite: 6]
- Connection = Y-connected[cite: 6]
- Full load speed = 125 rpm[cite: 6] (Assumed typographical error; corrected to 1425 rpm for meaningful analysis).
- Rotor impedance = $0.4 + j4 \, \Omega$ [cite: 6]
- Rotor/stator turn ratio $K = 0.8$ [cite: 6]
- Windage and friction losses = 500 W[cite: 6]

Synchronous speed $N_s = 1500 \, \text{rpm}$.

Operating slip $s = \frac{1500 - 1425}{1500} = 0.05$.

Stator phase voltage $V_1 = \frac{440}{\sqrt{3}} = 254.03 \, \text{V}$.

Rotor induced EMF at standstill $E_2 = K \cdot V_1 = 0.8 \times 254.03 = 203.23 \, \text{V}$.

Synchronous angular velocity $\omega_s = \frac{2\pi \times 1500}{60} = 157.08 \, \text{rad/s}$.

(ii) Rotor current and full load rotor Cu-loss:

$$I_2 = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}} = \frac{0.05 \times 203.23}{\sqrt{0.4^2 + (0.05 \times 4)^2}} = \frac{10.16}{\sqrt{0.16 + 0.04}} = 22.72 \, \text{A}$$

$$\text{Full load rotor Cu-loss} = 3I_2^2 R_2 = 3 \times (22.72)^2 \times 0.4 = 619.4 \, \text{W}$$

(iii) Power output:

$$\text{Air-gap power } P_{ag} = \frac{\text{Rotor Cu-loss}}{s} = \frac{619.4}{0.05} = 12388 \, \text{W}$$

$$\text{Mechanical power developed } P_d = P_{ag}(1 - s) = 12388(1 - 0.05) = 11768.6 \, \text{W}$$

$$P_{out} = P_d - P_{fw} = 11768.6 - 500 = 11268.6 \text{ W (or 11.27 kW)}$$

(i) Full load torque (T_{fl}):

$$T_{fl} = \frac{P_{ag}}{\omega_s} = \frac{12388}{157.08} = 78.86 \text{ N-m}$$

(iv) Maximum torque and speed at which it occurs:

$$s_{mT} = \frac{R_2}{X_2} = \frac{0.4}{4} = 0.1$$

$$\text{Speed at } T_{max} = N_s(1 - s_{mT}) = 1500(1 - 0.1) = 1350 \text{ rpm}$$

$$T_{max} = \frac{3E_2^2}{2\omega_s X_2} = \frac{3 \times (203.23)^2}{2 \times 157.08 \times 4} = 98.63 \text{ N-m}$$

(v) Starting current (I_{st}):

$$I_{st} = \frac{E_2}{\sqrt{R_2^2 + X_2^2}} = \frac{203.23}{\sqrt{0.4^2 + 4^2}} = 50.55 \text{ A}$$

(vi) Starting torque (T_{st}):

$$T_{st} = \frac{3}{\omega_s} \cdot I_{st}^2 \cdot R_2 = \frac{3}{157.08} \times (50.55)^2 \times 0.4 = 19.52 \text{ N-m}$$

Question 4

Thought Process

Torque is proportional to the square of the voltage and depends on the slip: $T \propto V_1^2 \frac{sR_2}{R_2^2 + (sX_2)^2}$. I will calculate the torque proportionality constant at the initial full-load slip. Then, I will find the new slip corresponding to "half full-load speed". By equating the torque at the new slip (with a new voltage V_2) to the original full-load torque, I can find the ratio $(V_2/V_1)^2$ and hence the percentage voltage reduction. The power factor will be computed using the rotor impedance triangle at the new slip.

Answer:

Given Data:

- Rotor resistance $R_2 = 0.015 \text{ } \Omega/\text{phase}$ [cite: 6]
- Standstill reactance $X_2 = 0.09 \text{ } \Omega/\text{phase}$ [cite: 6]
- Full load slip at normal voltage $s_1 = 3\% = 0.03$ [cite: 6]

Initial full load torque proportionality at voltage V_1 :

$$T_{fl} \propto V_1^2 \frac{s_1 R_2}{R_2^2 + (s_1 X_2)^2} = V_1^2 \frac{0.03 \times 0.015}{0.015^2 + (0.03 \times 0.09)^2}$$

$$T_{fl} \propto V_1^2 \frac{0.00045}{0.000225 + 0.00000729} = 1.9372 \cdot V_1^2$$

Let N_s be the synchronous speed. Full-load speed $N_{fl} = N_s(1 - 0.03) = 0.97N_s$.

Half full-load speed $N_2 = 0.5 \times 0.97N_s = 0.485N_s$.

New slip at half speed (s_2):

$$s_2 = \frac{N_s - 0.485N_s}{N_s} = 0.515$$

Torque proportionality at the new voltage V_2 :

$$T_2 \propto V_2^2 \frac{0.515 \times 0.015}{0.015^2 + (0.515 \times 0.09)^2} = V_2^2 \frac{0.007725}{0.000225 + 0.002148} = 3.255 \cdot V_2^2$$

To develop full load torque, set $T_2 = T_{fl}$:

$$3.255 \cdot V_2^2 = 1.9372 \cdot V_1^2 \implies \left(\frac{V_2}{V_1}\right)^2 = \frac{1.9372}{3.255} = 0.5951 \implies \frac{V_2}{V_1} = 0.7714$$

Percentage reduction in stator voltage = $(1 - 0.7714) \times 100\% = 22.86\%$.

Rotor power factor at the new slip s_2 :

$$\cos \phi_2 = \frac{R_2}{\sqrt{R_2^2 + (s_2 X_2)^2}} = \frac{0.015}{\sqrt{0.015^2 + (0.515 \times 0.09)^2}} = \frac{0.015}{0.0487} = 0.308 \text{ lagging}$$

Question 5

Thought Process

The power output is given. Adding the friction and windage losses yields the gross mechanical power developed (P_d). The rotor input (air-gap power, P_{ag}) is related to P_d by the factor $1/(1-s)$. The rotor copper loss is simply $s \cdot P_{ag}$. The shaft torque and gross electromagnetic torque are obtained by dividing the respective powers (P_{out} and P_{ag}) by their corresponding angular velocities (ω_r and ω_s).

Answer:

Given Data:

- Output power $P_{out} = 18.65 \text{ kW} = 18650 \text{ W}$ [cite: 6]
- Number of poles $P = 4$ [cite: 6]
- Frequency $f = 50 \text{ Hz}$ [cite: 6]
- Friction and windage losses = 2.5% of output[cite: 6]
- Full-load slip $s = 4\% = 0.04$ [cite: 6]

Friction and windage losses $P_{fw} = 0.025 \times 18650 = 466.25 \text{ W}$.

Gross mechanical power developed $P_d = P_{out} + P_{fw} = 18650 + 466.25 = 19116.25 \text{ W}$.

(b) The rotor input (P_{ag}):

$$P_{ag} = \frac{P_d}{1-s} = \frac{19116.25}{1-0.04} = \frac{19116.25}{0.96} = 19912.76 \text{ W}$$

(a) The rotor Cu loss (P_{rc}):

$$P_{rc} = s \cdot P_{ag} = 0.04 \times 19912.76 = 796.51 \text{ W}$$

(c) The shaft torque (T_{sh}):

Synchronous speed $N_s = \frac{120 \times 50}{4} = 1500 \text{ rpm}$.

Rotor speed $N_r = 1500(1 - 0.04) = 1440 \text{ rpm}$.

Rotor angular velocity $\omega_r = \frac{2\pi \times 1440}{60} = 150.8 \text{ rad/s}$.

$$T_{sh} = \frac{P_{out}}{\omega_r} = \frac{18650}{150.8} = 123.67 \text{ N-m}$$

(d) The gross electromagnetic torque (T_{em}):

Synchronous angular velocity $\omega_s = \frac{2\pi \times 1500}{60} = 157.08 \text{ rad/s}$.

$$T_{em} = \frac{P_{ag}}{\omega_s} = \frac{19912.76}{157.08} = 126.77 \text{ N-m}$$

Question 6

Thought Process

First, find the rotor phase voltage using the stator-to-rotor turns ratio. Calculate s_{mT} using R_2 and X_2 . Use the standard maximum torque formula $T_{max} = \frac{3E_2^2}{2\omega_s X_2}$. Given $T_{max} = 2T_{fl}$, apply the torque ratio equation $\frac{T_{fl}}{T_{max}} = \frac{2 \cdot s \cdot s_{mT}}{s^2 + s_{mT}^2} = 0.5$ to solve the quadratic for the full-load slip s . Finally, use the resulting torque and operating speed to calculate the gross power output in watts.

Answer:

Given Data:

- Line voltage = 400 V[cite: 6]
- Number of poles $P = 4$ [cite: 6]
- Frequency $f = 50 \text{ Hz}$ [cite: 6]
- Rotor resistance $R_2 = 0.01 \Omega/\text{phase}$ [cite: 6]
- Rotor reactance $X_2 = 0.1 \Omega/\text{phase}$ [cite: 6]
- $T_{max} = 2 \cdot T_{fl}$ [cite: 6]
- Ratio of stator to rotor turns $K' = 4$ [cite: 6]

Stator phase voltage $V_1 = \frac{400}{\sqrt{3}} = 230.94 \text{ V}$.

Rotor induced EMF/phase $E_2 = \frac{V_1}{4} = \frac{230.94}{4} = 57.735 \text{ V}$.

Synchronous speed $N_s = 1500 \text{ rpm} \implies \omega_s = \frac{2\pi \times 1500}{60} = 157.08 \text{ rad/s}$.

(a) Maximum torque and corresponding slip:

$$s_{mT} = \frac{R_2}{X_2} = \frac{0.01}{0.1} = 0.1$$

$$T_{max} = \frac{3E_2^2}{2\omega_s X_2} = \frac{3 \times (57.735)^2}{2 \times 157.08 \times 0.1} = \frac{10000}{31.416} = 318.3 \text{ N-m}$$

(b) Full load slip and power output: Using the torque ratio formula:

$$\frac{T_{fl}}{T_{max}} = \frac{2 \cdot s \cdot s_{mT}}{s^2 + s_{mT}^2} \implies 0.5 = \frac{2 \cdot s \cdot (0.1)}{s^2 + 0.01}$$

$$s^2 + 0.01 = 0.4s \implies s^2 - 0.4s + 0.01 = 0$$

Solving for full-load slip (s):

$$s = \frac{0.4 \pm \sqrt{0.16 - 0.04}}{2} = 0.2 \pm \sqrt{0.03} = 0.2 \pm 0.1732$$

The operating slip is the lower value: $s = 0.0268$.

Full-load torque $T_{fl} = \frac{T_{max}}{2} = \frac{318.3}{2} = 159.15 \text{ N-m}$.

Operating angular velocity $\omega_r = \omega_s(1 - s) = 157.08(1 - 0.0268) = 152.87 \text{ rad/s}$.

Power output (gross) $P_d = T_{fl} \cdot \omega_r = 159.15 \times 152.87 = 24329 \text{ W}$.

Question 7

Thought Process

The equivalent load resistance in an induction motor circuit is modeled as $R_L = R_2 \frac{1-s}{s}$. I will calculate this using the given slip and rotor resistance. Since the gross power output is given, the power output per phase is $9000/3 = 3000$ W. Using $P = I_2^2 R_L$, I can solve for the rotor current I_2 . Finally, the equivalent load voltage is $V_L = I_2 R_L$.

Answer:

Given Data:

- Slip at max torque s_{mT} ... wait, the problem asks for values "at this slip" meaning $s = 12\% = 0.12$. [cite: 6]
- Equivalent secondary resistance $R_2 = 0.08 \Omega/\text{phase}$ [cite: 6]
- Gross power output $P_d = 9000$ W [cite: 6]

Equivalent load resistance R_L :

$$R_L = R_2 \left(\frac{1-s}{s} \right) = 0.08 \left(\frac{1-0.12}{0.12} \right) = 0.08 \times \frac{0.88}{0.12} = 0.5867 \Omega/\text{phase}$$

Gross power output per phase $P_{d(ph)} = \frac{9000}{3} = 3000$ W.

Rotor current I_2 at this slip:

$$P_{d(ph)} = I_2^2 R_L \implies 3000 = I_2^2 \times 0.5867$$

$$I_2 = \sqrt{\frac{3000}{0.5867}} = \sqrt{5113.3} = 71.51 \text{ A}$$

Equivalent load voltage V_L :

$$V_L = I_2 R_L = 71.51 \times 0.5867 = 41.95 \text{ V}$$

Question 8

Thought Process

To find the full-load current, I will use the output power, efficiency, and power factor. In a standard starter problem, "blocked rotor current at 400 V is 40 A" refers to the phase current when the full voltage is applied across the winding. When started with a star-delta starter, the windings are connected in star, reducing the phase voltage by a factor of $\sqrt{3}$. Consequently, the starting line current will be the phase current divided by $\sqrt{3}$. The ratio is then $I_{st(line)}/I_{fl(line)}$.

Answer:

Given Data:

- Output power $P_{out} = 10 \text{ kW} = 10000$ W [cite: 6]
- Line voltage $V_L = 400$ V [cite: 6]
- Full load power factor $\cos \phi = 0.85$ [cite: 6]
- Full load efficiency $\eta = 0.88$ [cite: 6]
- Blocked rotor current at 400 V = 40 A [cite: 6]

Calculate input power (P_{in}):

$$P_{in} = \frac{P_{out}}{\eta} = \frac{10000}{0.88} = 11363.6 \text{ W}$$

Calculate full load line current (I_{fl}):

$$I_{fl} = \frac{P_{in}}{\sqrt{3}V_L \cos \phi} = \frac{11363.6}{\sqrt{3} \times 400 \times 0.85} = \frac{11363.6}{588.9} = 19.3 \text{ A}$$

The blocked rotor current of 40 A represents the short-circuit phase current ($I_{sc(ph)}$) at the full phase voltage (400 V in delta).

When started with a star-delta starter, the starting line current is equal to the starting phase current in star:

$$I_{st} = \frac{I_{sc(ph)}}{\sqrt{3}} = \frac{40}{\sqrt{3}} = 23.09 \text{ A}$$

Ratio of starting to full load current:

$$\text{Ratio} = \frac{I_{st}}{I_{fl}} = \frac{23.09}{19.3} = 1.196$$

Question 9

Thought Process

A star-delta starter limits the starting line current and torque to 1/3 of their respective Direct On-Line (DOL) values. Using the provided star-delta percentages, I will first calculate the hypothetical DOL starting current and torque. Then, I will apply the formulas for an auto-transformer starter, where starting current and torque scale by the square of the tapping fraction (x^2).

Answer:

Given Data:

- Star-delta starting current $I_{st,Y} = 180\%$ of $I_{fl} = 1.8I_{fl}$ [cite: 6]
- Star-delta starting torque $T_{st,Y} = 35\%$ of $T_{fl} = 0.35T_{fl}$ [cite: 6]
- Auto-transformer tapping $x = 75\% = 0.75$ [cite: 6]

Determine the equivalent Direct On-Line (DOL) parameters:

$$I_{st,DOL} = 3 \times I_{st,Y} = 3 \times 1.8I_{fl} = 5.4I_{fl}$$

$$T_{st,DOL} = 3 \times T_{st,Y} = 3 \times 0.35T_{fl} = 1.05T_{fl}$$

Now, apply the auto-transformer scaling factor ($x^2 = 0.75^2 = 0.5625$):

Starting current with auto-transformer ($I_{st,auto}$):

$$I_{st,auto} = x^2 \cdot I_{st,DOL} = 0.5625 \times 5.4I_{fl} = 3.0375I_{fl} \text{ (or 303.75\% of full load current)}$$

Starting torque with auto-transformer ($T_{st,auto}$):

$$T_{st,auto} = x^2 \cdot T_{st,DOL} = 0.5625 \times 1.05T_{fl} = 0.5906T_{fl} \text{ (or 59.06\% of full load torque)}$$

Question 10

Thought Process

Torque produced by a single cage is proportional to the air-gap power transferred to it, given by $T \propto I_2^2 \frac{R_2}{s} = \frac{V^2}{|R_2/s + jX_2|^2} \frac{R_2}{s}$. I will calculate this proportionality for both the inner and outer cages at standstill ($s = 1$) and at 5% slip ($s = 0.05$). The ratio is simply T_{outer}/T_{inner} .

Answer:

Given Data:

- Inner cage impedance at standstill $Z_i = 0.4 + j2 \Omega$ ($R_i = 0.4, X_i = 2$)[cite: 6]
- Outer cage impedance at standstill $Z_o = 2 + j0.4 \Omega$ ($R_o = 2, X_o = 0.4$)[cite: 6]

(i) At standstill ($s = 1$):

$$T_i \propto \frac{R_i}{R_i^2 + X_i^2} = \frac{0.4}{0.4^2 + 2^2} = \frac{0.4}{0.16 + 4} = 0.09615$$

$$T_o \propto \frac{R_o}{R_o^2 + X_o^2} = \frac{2}{2^2 + 0.4^2} = \frac{2}{4 + 0.16} = 0.4807$$

Ratio of torques (Outer to Inner) at standstill:

$$\frac{T_o}{T_i} = \frac{0.4807}{0.09615} = 5$$

(ii) At 5% slip ($s = 0.05$):

$$T_i \propto \frac{R_i/s}{(R_i/s)^2 + X_i^2} = \frac{0.4/0.05}{(0.4/0.05)^2 + 2^2} = \frac{8}{8^2 + 4} = \frac{8}{68} = 0.1176$$

$$T_o \propto \frac{R_o/s}{(R_o/s)^2 + X_o^2} = \frac{2/0.05}{(2/0.05)^2 + 0.4^2} = \frac{40}{40^2 + 0.16} = \frac{40}{1600.16} = 0.02499$$

Ratio of torques (Outer to Inner) at 5% slip:

$$\frac{T_o}{T_i} = \frac{0.02499}{0.1176} = 0.2125$$

Question 11

Thought Process

The torque from each cage is given by $T \propto \frac{s \cdot R_2}{R_2^2 + (sX_2)^2}$. By setting the torque expressions for the inner and outer cages equal to each other, I can eliminate the common s and voltage terms, reducing the relation to an algebraic equation involving s^2 . I will plug in the given impedance values and solve for s .

Answer:

Given Data:

- Outer cage impedance $Z_o = 2 + j1.2 \Omega$ ($R_o = 2, X_o = 1.2$)[cite: 6]
- Inner cage impedance $Z_i = 0.5 + j3.5 \Omega$ ($R_i = 0.5, X_i = 3.5$)[cite: 6]

Set the torques equal:

$$\frac{s \cdot R_o}{R_o^2 + s^2 X_o^2} = \frac{s \cdot R_i}{R_i^2 + s^2 X_i^2}$$

Assuming $s \neq 0$, cancel s and substitute the values:

$$\frac{2}{2^2 + s^2(1.2)^2} = \frac{0.5}{0.5^2 + s^2(3.5)^2}$$

$$\frac{2}{4 + 1.44s^2} = \frac{0.5}{0.25 + 12.25s^2}$$

Cross-multiply to solve for s :

$$2(0.25 + 12.25s^2) = 0.5(4 + 1.44s^2)$$

$$0.5 + 24.5s^2 = 2 + 0.72s^2$$

Collect the s^2 terms:

$$(24.5 - 0.72)s^2 = 2 - 0.5$$

$$23.78s^2 = 1.5 \implies s^2 = \frac{1.5}{23.78} = 0.063078$$

$$s = \sqrt{0.063078} = 0.2511$$

The slip at which the two cages develop equal torques is 25.11%.